Task 2: Loan Approval Prediction Report

1. Executive Summary

This report presents a supervised machine learning analysis for predicting loan approval status based on borrower and loan-related information. The objective of this task was to build a model that can predict whether a loan application is likely to be approved or rejected.

The dataset contained applicant details such as gender, marital status, dependents, education, employment status, applicant income, co-applicant income, loan amount, loan amount term, credit history, property area, and loan approval status.

The analysis included data preprocessing, missing value handling, feature engineering, categorical encoding, feature scaling, class imbalance handling, model training, model evaluation, and threshold tuning. Three models were compared: Logistic Regression, Decision Tree, and Random Forest.

Random Forest performed the best based on F1-score. It achieved an accuracy of **0.8699**, precision of **0.8710**, recall of **0.9529**, F1-score of **0.9101**, and ROC-AUC of **0.8550**. Although Logistic Regression achieved a slightly higher ROC-AUC score, Random Forest provided better recall and F1-score, making it more suitable for this loan approval prediction task.

Based on threshold tuning, a threshold of **0.50** is recommended for initial deployment because it achieved the best F1-score. If the business wants a more conservative approval strategy, a threshold of **0.60** can be considered because it improves precision, but it slightly reduces recall.

2. Objective

The objective of this task is to build a supervised machine learning model to predict loan approval status using borrower and loan-related features.

The main focus areas of this task are:

- Handling missing values
- Encoding categorical variables
- Scaling numerical features
- Handling class imbalance
- Comparing multiple machine learning models
- Evaluating models using precision, recall, F1-score, and ROC-AUC
- Performing threshold tuning
- Providing business-oriented interpretation of model results

3. Dataset Overview

The dataset contains **614 records** and **13 columns**. The target column is **Loan_Status**, where:

- **Y** represents loan approved
- **N** represents loan not approved

Main Features

Feature	**Description**
Loan_ID	Unique loan application ID
Gender	Gender of the applicant
Married	Marital status of the applicant
Dependents	Number of dependents
Education	Education status of the applicant
Self_Employed	Whether the applicant is self-employed
ApplicantIncome	Income of the main applicant
CoapplicantIncome	Income of the co-applicant
LoanAmount	Loan amount requested
Loan_Amount_Term	Term duration of the loan
Credit_History	Credit history of the applicant
Property_Area	Area type of the property
Loan_Status	Final loan approval status

4. Tools Used

The following tools and technologies were used for this task:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Google Colab

- GitHub

5. Data Preprocessing

The raw dataset contained missing values in multiple columns. Before model training, the data was cleaned and prepared using the following steps:

Missing Value Handling

Categorical missing values were filled using the **mode**, because mode represents the most frequent category in each column.

Columns handled using mode:

- Gender
- Married
- Dependents
- Self_Employed
- Credit_History
- Loan_Amount_Term

Numerical missing values in **LoanAmount** were filled using the **median**, because median is more stable when data contains outliers.

Feature Engineering

A new feature called **Total_Income** was created by combining applicant income and co-applicant income.

Total_Income = ApplicantIncome + CoapplicantIncome

Log transformation was applied to reduce skewness in income and loan amount features:

LoanAmount_Log = log(LoanAmount)

Total_Income_Log = log(Total_Income)

Encoding

Categorical variables were converted into numerical format using one-hot encoding.

Feature Scaling

StandardScaler was used to scale features before model training. Scaling was especially useful for Logistic Regression because it is sensitive to feature magnitude.

Class Imbalance Handling

The target classes were imbalanced, so class imbalance was handled using:

class_weight = "balanced"

This allowed the models to give more balanced attention to both approved and rejected loan applications.

6. Models Compared

Three supervised machine learning models were trained and compared:

1. **Logistic Regression**
   A simple and interpretable linear model useful for binary classification.
2. **Decision Tree**
   A tree-based model that can capture non-linear patterns but may overfit.
3. **Random Forest**
   An ensemble model that combines multiple decision trees and generally provides better stability and performance.

7. Model Performance

The models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC.

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.8293	0.8721	0.8824	0.8772	0.8632
Decision Tree	0.7642	0.8590	0.7882	0.8221	0.7494
Random Forest	0.8699	0.8710	0.9529	0.9101	0.8550

Model Selection

Random Forest was selected as the best model because it achieved the highest:

- Accuracy: **0.8699**
- Recall: **0.9529**
- F1-score: **0.9101**

Although Logistic Regression achieved a slightly higher ROC-AUC score of **0.8632**, Random Forest gave better recall and F1-score. Since loan approval prediction requires balancing approval opportunities and risk control, F1-score was considered an important metric.

8. Threshold Tuning

Threshold tuning was performed to understand how different probability thresholds affect precision, recall, and F1-score.

Threshold	Precision	Recall	F1 Score
0.30	0.8058	0.9765	0.8830
0.40	0.8300	0.9765	0.8973
0.45	0.8454	0.9647	0.9011
0.50	0.8710	0.9529	0.9101
0.60	0.8953	0.9059	0.9006
0.70	0.9130	0.7412	0.8182

Suggested Deployment Threshold

The threshold of **0.50** is recommended for initial deployment because it achieved the best F1-score of **0.9101**, with precision of **0.8710** and recall of **0.9529**.

If the business wants to reduce risky approvals and follow a more conservative approval strategy, a threshold of **0.60** can be considered. At this threshold, precision improves to **0.8953**, but recall decreases to **0.9059**.

This means:

- Lower threshold → approves more applicants but may increase risk
- Higher threshold → safer approvals but may reject some eligible applicants
- 0.50 threshold → best balance in this analysis

9. Business Interpretation

The model can help financial institutions identify applicants who are more likely to receive loan approval based on historical borrower information.

Random Forest performed well because it captured relationships between borrower features and approval status. The model showed strong recall, meaning it was effective in identifying approved loan applications.

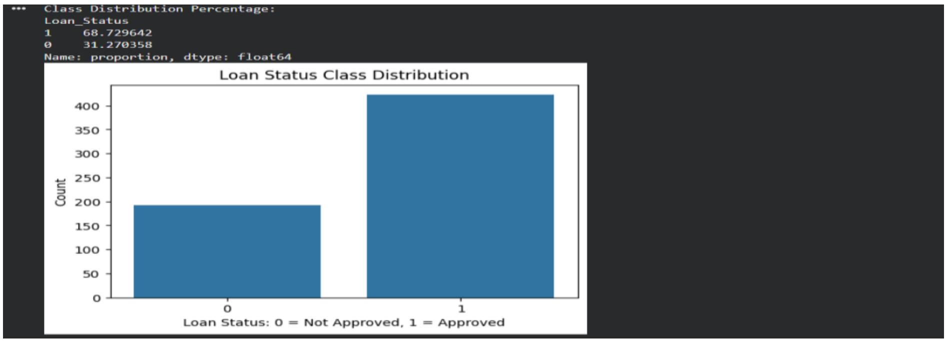
From a business perspective:

- High recall helps reduce the chance of rejecting eligible applicants.
- High precision helps reduce the chance of approving risky applicants.
- A balanced F1-score is important because loan approval involves both customer opportunity and financial risk.
- The model should support loan officers but should not fully replace human review.

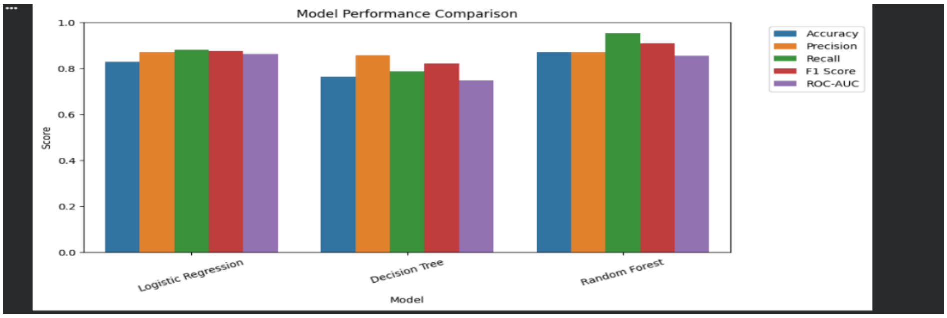
The final decision should still include business rules, credit policy, compliance checks, and manual review where necessary.

10. Key Charts

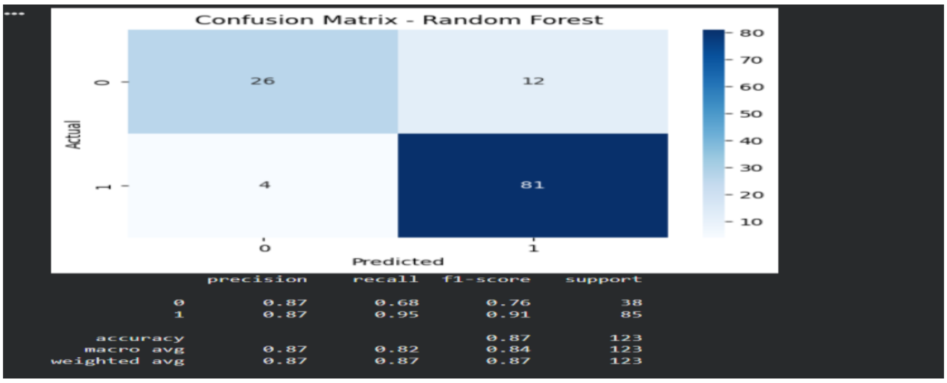
10.1 Loan Status Distribution



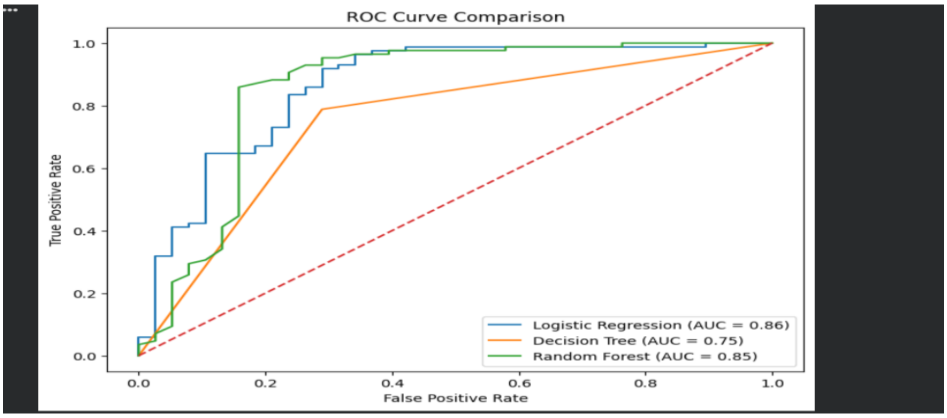
10.2 Model Performance Comparison



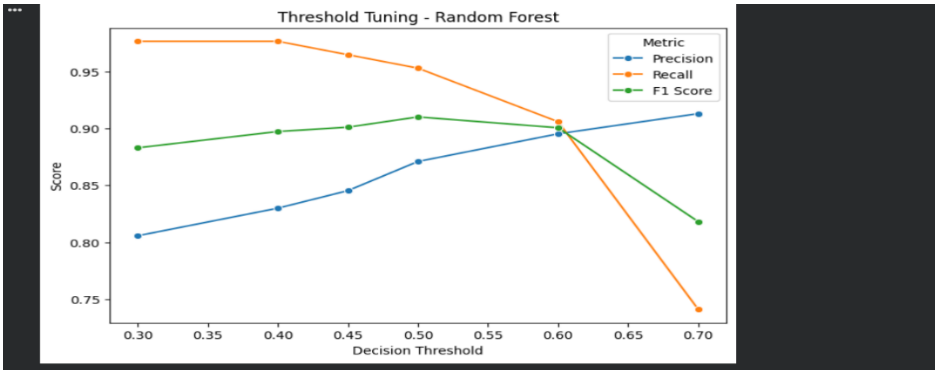
10.3 Confusion Matrix - Random Forest



10.4 ROC Curve Comparison



10.5 Threshold Tuning Chart



10.6 Business Interpretation Output

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BUSINESS INTERPRETATION
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The best performing model based on F1 Score is: Random Forest
For loan approval prediction, precision and recall must be balanced carefully.
Higher precision reduces the risk of approving applicants who may default.
Higher recall helps identify more eligible applicants who should be approved.
A threshold around 0.45 to 0.50 can be considered depending on the business goal.
If the company wants safer approvals, use a higher threshold.
If the company wants to approve more eligible customers, use a slightly lower threshold.
The model should support human decision-making and should not fully replace manual loan review.
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11. Limitation

The dataset contains only **614 records**, which is relatively small for a real-world loan approval system. The model performance may change if trained on a larger and more recent dataset.

The dataset also contains limited financial and risk-related information. In a real banking or financial environment, additional factors such as credit score, repayment history, debt-to-income ratio, employment stability, existing liabilities, and policy rules would be important.

Therefore, this model should be treated as a learning and decision-support model, not a final production-ready loan approval system.

12. GitHub / Notebook Link

GitHub Task Folder:
https://github.com/saitharun-student-27/InternSpark-Data-Science-Internship/tree/main/Task-3-Loan-Approval-Prediction

Notebook Link:

https://colab.research.google.com/drive/1vGblan1hQKq-fK7hIl6uHH0jv24SdNmD?usp=sharing

13. Conclusion

This task helped build an end-to-end supervised machine learning pipeline for loan approval prediction. The work included data preprocessing, missing value handling, feature engineering, encoding, scaling, class imbalance handling, model comparison, threshold tuning, and business interpretation.

Random Forest was selected as the final model because it achieved the best F1-score and recall among the tested models. A threshold of **0.50** is recommended for initial deployment because it provides the best balance between precision and recall.

The main learning from this task was that model building is not only about achieving high accuracy. It is also about understanding trade-offs, selecting the right evaluation metrics, handling class imbalance, and interpreting results in a business context.